

STUDYHUB
CLASS 10 — MP BOARD
MATHEMATICS (STANDARD)
SAMPLE QUESTION PAPER — SET 1

Time: 3 Hours	Maximum Marks: 75
Medium: English	Set: 1

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Questions 1 to 5 are objective-type questions.
3. Internal choices are given in Questions 6 to 23.
4. Use of a calculator is not permitted.
5. Draw neat and clean diagrams wherever required.
6. Show necessary steps in numerical questions.

PART A — OBJECTIVE TYPE QUESTIONS

Q.1 Choose the correct alternative. [6 Marks]

(1) If HCF of 84 and 126 is 42, then their LCM is:

- A. 168
- B. 210
- C. 252
- D. 336

(2) The zeroes of the polynomial $x^2 - 7x + 12$ are:

- A. 2 and 6
- B. 3 and 4
- C. 1 and 12
- D. -3 and -4

(3) The pair of equations $2x + 3y = 7$ and $4x + 6y = 14$ has:

- A. No solution
- B. Unique solution
- C. Infinitely many solutions
- D. Exactly two solutions

(4) The common difference of the AP 11, 8, 5, 2, ... is:

- A. 3
- B. -3
- C. 5
- D. -5

(5) If $\tan \theta = 1$, where $0^\circ < \theta < 90^\circ$, then θ is:

- A. 30°
- B. 45°
- C. 60°
- D. 90°

(6) The probability of an impossible event is:

- A. 0
- B. 1
- C. $1/2$
- D. -1

Q.2 Fill in the blanks. [6 Marks]

(1) The decimal expansion of a rational number whose denominator has only 2 and 5 as prime factors is _____.

(2) The degree of a non-zero constant polynomial is _____.

(3) The distance of the point (0, -5) from the x-axis is _____ units.

(4) The 10th term of the AP 3, 7, 11, 15, ... is _____.

(5) $\sin^2 \theta + \cos^2 \theta =$ _____.

(6) The probability of a sure event is _____.

Q.3 Write True or False. [6 Marks]

- (1) Every irrational number has a non-terminating, non-repeating decimal expansion.
- (2) A quadratic polynomial can have three distinct zeroes.
- (3) The graph of two intersecting straight lines represents a pair of equations having a unique solution.
- (4) The sum of the first n terms of an AP is always equal to its n th term.
- (5) $\sec^2\theta - \tan^2\theta = 1$.
- (6) The radius drawn to the point of contact of a tangent is perpendicular to the tangent.

Q.4 Match the following. [6 Marks]

Column A	Column B
(i) $x^2 - 9$	(a) 2
(ii) Distance between (0,0) and (3,4)	(b) $x=3, -3$
(iii) Common difference of 5,9,13,17,...	(c) 5
(iv) $\cos 60^\circ$	(d) $1/2$
(v) Tangent to a circle	(e) One point of contact
(vi) Probability of getting an even number on a fair die	(f) $1/2$

Q.5 Answer in one word or one sentence. [6 Marks]

- (1) State Euclid's division lemma.
- (2) What is the sum of zeroes of ax^2+bx+c ?
- (3) What are the coordinates of the origin?
- (4) Write the formula for the n th term of an AP.
- (5) What is the value of $\sin 90^\circ$?
- (6) How many tangents can be drawn to a circle from a point lying outside it?

PART B — SHORT ANSWER QUESTIONS

Q.6 Using Euclid's division algorithm, find the HCF of 135 and 225. [2 Marks]

OR

Show that $\sqrt{7}$ is irrational. [2 Marks]

Q.7 Find the zeroes of the polynomial $x^2 - 5x + 6$ and verify the relationship between its zeroes and coefficients. [2 Marks]

OR

If one zero of the polynomial $2x^2 + 5x + k$ is 1, find the value of k . [2 Marks]

Q.8 Solve the following pair of equations: $x+y=9$ and $2x-y=6$. [2 Marks]

OR

Determine whether the pair $3x+2y=8$ and $6x+4y=16$ has a unique solution. [2 Marks]

Q.9 Find the 18th term of the AP 7,12,17,22,... [2 Marks]

OR

Which term of the AP 4,9,14,19,... is 79? [2 Marks]

Q.10 Find the distance between A(2,3) and B(8,11). [2 Marks]

OR

Find the coordinates of the midpoint of the line segment joining P(-4,7) and Q(6,-3). [2 Marks]

Q.11 Prove that $(1 - \cos^2\theta)/\sin^2\theta = 1$. [2 Marks]

OR

Evaluate $2\sin 30^\circ + \cos 60^\circ$. [2 Marks]

Q.12 From a point outside a circle, two tangents are drawn. If one tangent has length 8 cm, find the length of the other tangent. Give reason. [2 Marks]

OR

What is the angle between a tangent and the radius drawn to the point of contact? State the theorem used. [2 Marks]

Q.13 Find the area of a sector of a circle of radius 14 cm and angle 90° . Use $\pi = 22/7$. [2 Marks]

OR

Find the circumference of a circle whose area is 154 cm^2 . [2 Marks]

Q.14 Find the volume of a cylinder having radius 7 cm and height 10 cm. Use $\pi = 22/7$. [2 Marks]

OR

Find the total surface area of a cube whose side is 6 cm. [2 Marks]

Q.15 Find the mean of 8,12,15,10,5. [2 Marks]

OR

Find the median of 7,3,9,5,11. [2 Marks]

Q.16 A bag contains 5 red balls and 3 blue balls. One ball is drawn at random. Find the probability that the ball is blue. [2 Marks]

OR

A die is thrown once. Find the probability of getting a number greater than 4. [2 Marks]

Q.17 A ladder 13 m long reaches a window 12 m above the ground. Find the distance of the foot of the ladder from the wall. [2 Marks]

OR

If $\tan\theta = 3/4$, find $\sin\theta$ and $\cos\theta$, where θ is acute. [2 Marks]

PART C — 3-MARK QUESTIONS

Q.18 Solve the quadratic equation $2x^2 - 7x + 3 = 0$ using the quadratic formula. [3 Marks]

OR

Find the nature of the roots of $3x^2 - 4x + 5 = 0$ without solving the equation. [3 Marks]

Q.19 In $\triangle ABC$, $DE \parallel BC$, where D lies on AB and E lies on AC. If $AD = 3$ cm, $DB = 2$ cm and $AE = 4.5$ cm, find EC using the Basic Proportionality Theorem. [3 Marks]

OR

Prove that two triangles are similar if their corresponding angles are equal. State the criterion used. [3 Marks]

Q.20 The number of books read by students is: books = 0,1,2,3,4 and students = 3,7,10,6,4. Find the mean number of books. [3 Marks]

OR

A coin is tossed 200 times and heads appear 112 times. Find the experimental probability of getting heads and tails. [3 Marks]

PART D — LONG ANSWER QUESTIONS

Q.21 A two-digit number is four times the sum of its digits. If 18 is added to the number, its digits are reversed. Find the number. [4 Marks]

OR

The sum of two numbers is 27 and their product is 182. Find the numbers by forming a quadratic equation. [4 Marks]

Q.22 A solid is in the shape of a cone surmounted by a hemisphere. The common radius is 7 cm and the height of the cone is 24 cm. Find: (i) slant height, (ii) curved surface area of cone, (iii) curved surface area of hemisphere, (iv) total outer surface area. Use $\pi = 22/7$. [4 Marks]

OR

A cylindrical tank has radius 3.5 m and height 10 m. Find: (i) its capacity in litres, (ii) the area of metal sheet required to make the closed tank. Use $\pi = 22/7$. [4 Marks]

Q.23 From the top of a 20 m high building, the angle of elevation of the top of a tower is 30° , while the angle of depression of the foot of the tower is 45° . Find: (i) distance between the building and tower, (ii) height of the tower. [4 Marks]

OR

A point P is 25 cm away from the centre O of a circle of radius 7 cm. Tangents from P touch the circle at A and B. Find: (i) PA, (ii) PB, (iii) the area of quadrilateral OAPB. [4 Marks]

ANSWER KEY — SHORT

Q.1: C, B, C, B, B, A.

Q.2: Terminating; 0; 5; 39; 1; 1.

Q.3: True; False; True; False; True; True.

Q.4: (i)-(b), (ii)-(c), (iii)-(a), (iv)-(d), (v)-(e), (vi)-(f).

Q.5: $a = bq + r$, $0 \leq r < b$; $-b/a$; (0,0); $a_n = a + (n-1)d$; 1; 2.

Q.6 HCF=45. Q.7 Zeroes=2,3. Q.8 $x=5, y=4$. Q.9 18th term=92.

Q.10 Distance=10 units. Q.11=1. Q.12 Other tangent=8 cm.

Q.13 Sector area=154 cm². Q.14 Volume=1540 cm³. Q.15 Mean=10. Q.16 Probability=3/8. Q.17 Distance=5 m.

Q.18 Roots=3 and 1/2. Q.19 EC=3 cm. Q.20 Mean=2.05 books.

Q.21 Number=36. Q.22 Slant height=25 cm; total outer surface area=1375 cm².

Q.23 Distance=20 m; tower height= $20+20\sqrt{3}$ m.

MARKS VERIFICATION

Section	Questions	Marks
Objective	Q.1-5	30
Short Answer	Q.6-17	24
3-Mark	Q.18-20	9
Long Answer	Q.21-23	12
TOTAL	23 Questions	75

Final verification: $30 + 24 + 9 + 12 = 75$ Marks.